Assignment 2

https://github.com/mrisney/spatial-applications-assignment-two

Course: Spatial Applications IV (Public Health GIS), Spring 2025

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Characterizing Fast-Food Distribution in Baltimore City

Spatial Analysis of Fast-Food Locations at the Census Tract Level

Aims and Hypotheses

Aims:

- Perform exploratory spatial data analysis characterizing the spatial distribution of fast-food locations in Baltimore City at the Census tract level
- Focus on clustering, cluster detection, and spatial variation in the number of fast-food locations
- Analyze the relationship between fast-food locations and census-derived variables (total population and population below 25)
- Investigate potential clustering of fast-food locations around schools

Hypotheses:

- Fast-food locations in Baltimore City will show significant spatial clustering
- Significant clusters will be identifiable through spatial scan statistics
- Census variables (population size and age structure) will influence the distribution of clusters
- Fast-food locations will cluster around schools with possibly varying effects by school level

Methods

Spatial Data and GIS

- Fast-food data provided as point locations (Excel file) covering the Baltimore City MSA
- Baltimore City census tracts shapefile (2020 Census Tracts)
- Baltimore City boundary shapefile
- All shapefiles and point data reprojected to UTM Zone 18N (EPSG:26918)
- Fast-food points aggregated to census tracts within Baltimore City boundaries
- Census API used to obtain total population data from ACS 2020
- Coordinates exported in meters for spatial analysis
- Data outputs prepared in format suitable for SaTScan spatial analysis

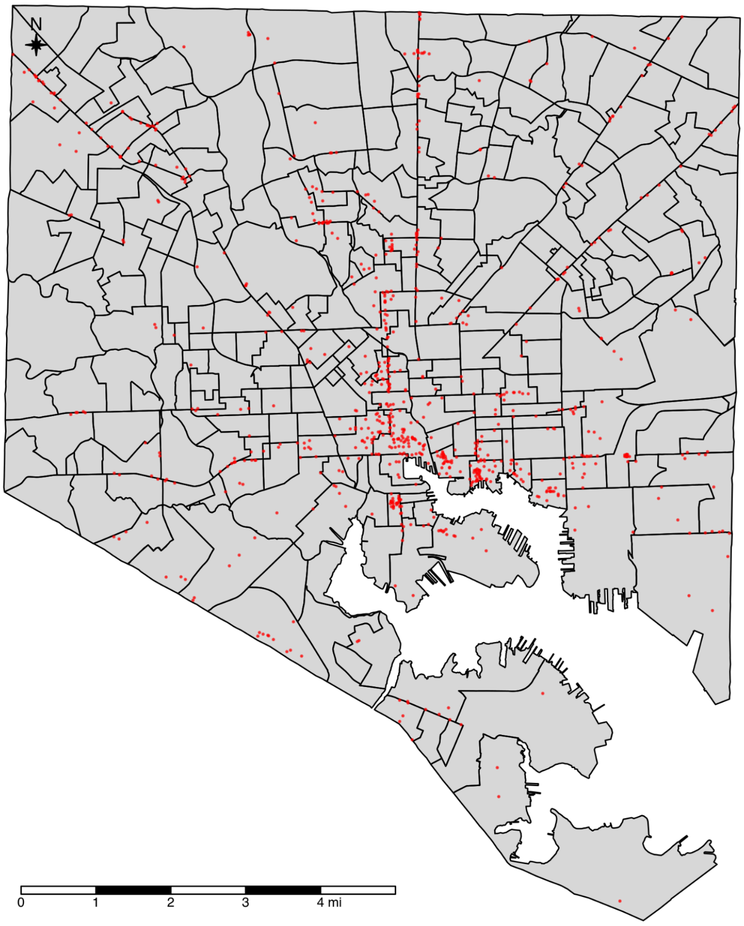
Spatial Statistics

- Global and Local Spatial Autocorrelation analysis (Moran's I) to assess clustering patterns
- Semivariogram analysis to quantify spatial dependence and determine range of influence
- Spherical variogram model fitted based on empirical semivariogram
- Purely spatial Poisson scan using SaTScan to identify statistically significant clusters
- Cross-validation performed to assess model accuracy (RMSE and MAE metrics)
- Statistical comparison of fast-food density within and outside school buffer zones
- Correlation analysis of distance to schools and fast-food density

Results

1. Fast-Food Point Distribution

Fast-Food Locations in Baltimore City

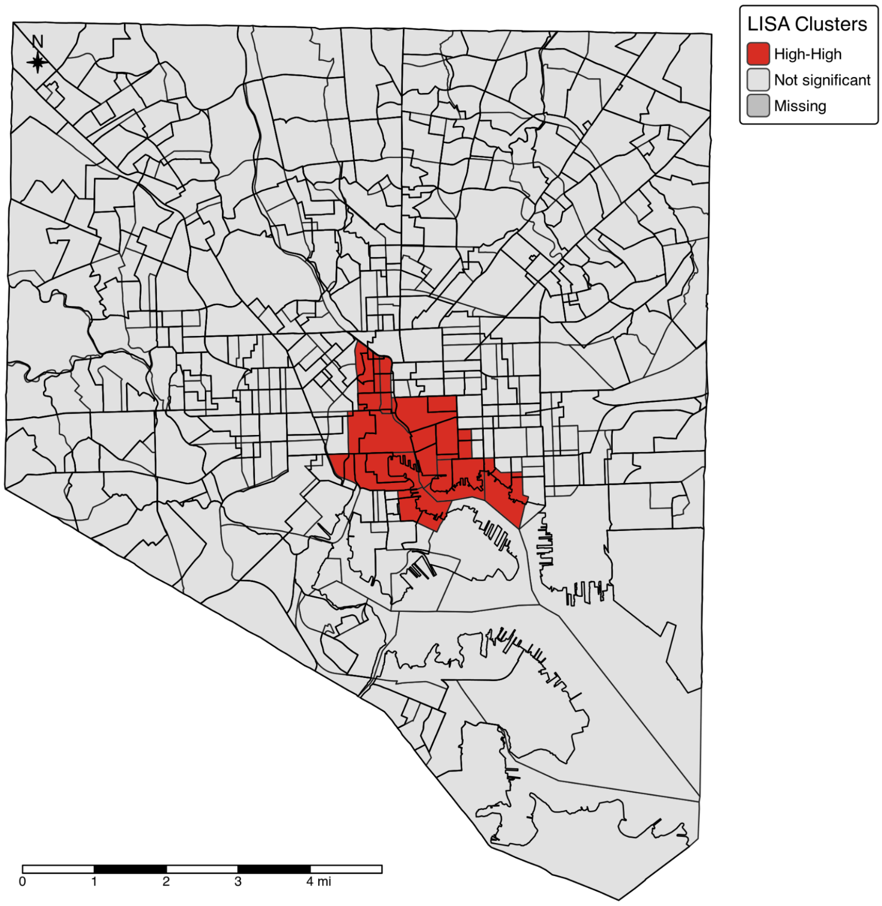


- Fast-food locations in Baltimore City show an uneven spatial distribution
- Points cluster in central-north and eastern areas, with gaps in western and southeastern regions
- Total of **823** unique fast-food establishments identified within city limits

Number of census tracts analyzed: ***199***
Mean fast-food locations per tract: ***4.14***
Maximum fast-food locations in a tract: ***84***

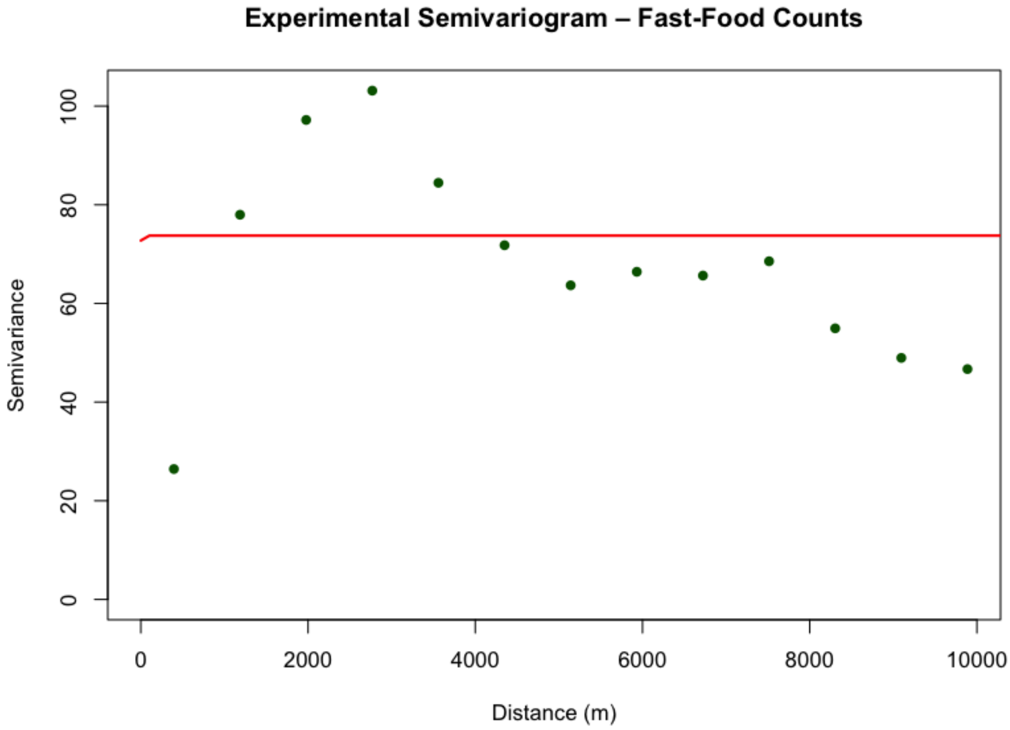
2. Spatial Autocorrelation Analysis

Local Moran's I Clusters of Fast-Food Counts
Baltimore City, MD



- Global Moran's I value of 0.2615 ($p < 0.001$) indicates significant positive spatial autocorrelation
- Expected value of -0.0051 with variance of 0.0012 confirms pattern deviates from spatial randomness
- High-High clusters predominantly located in central-eastern and western corridors of the city
- Pattern suggests fast-food establishments tend to concentrate in specific Baltimore neighborhood

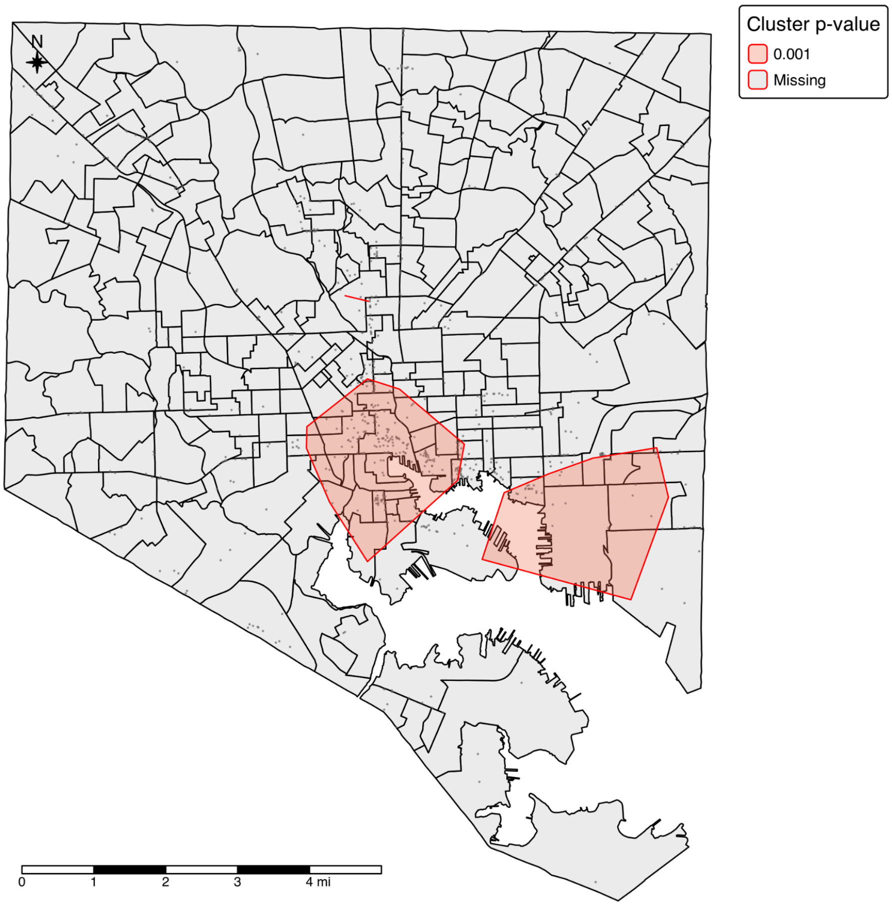
3. Semivariogram Analysis



- Experimental semivariogram shows spatial dependence up to approximately 6856 meters (4.3 miles)
- Spherical model provided best fit with nugget of 73.75, partial sill of 0, and range of 6856 meters (4.3 miles)
- Model indicates pure nugget effect spatial structure (nugget-to-sill ratio of 1.0)
- Pattern suggests high local variability with spatial correlation extending up to ~7 km (~4.3 miles)

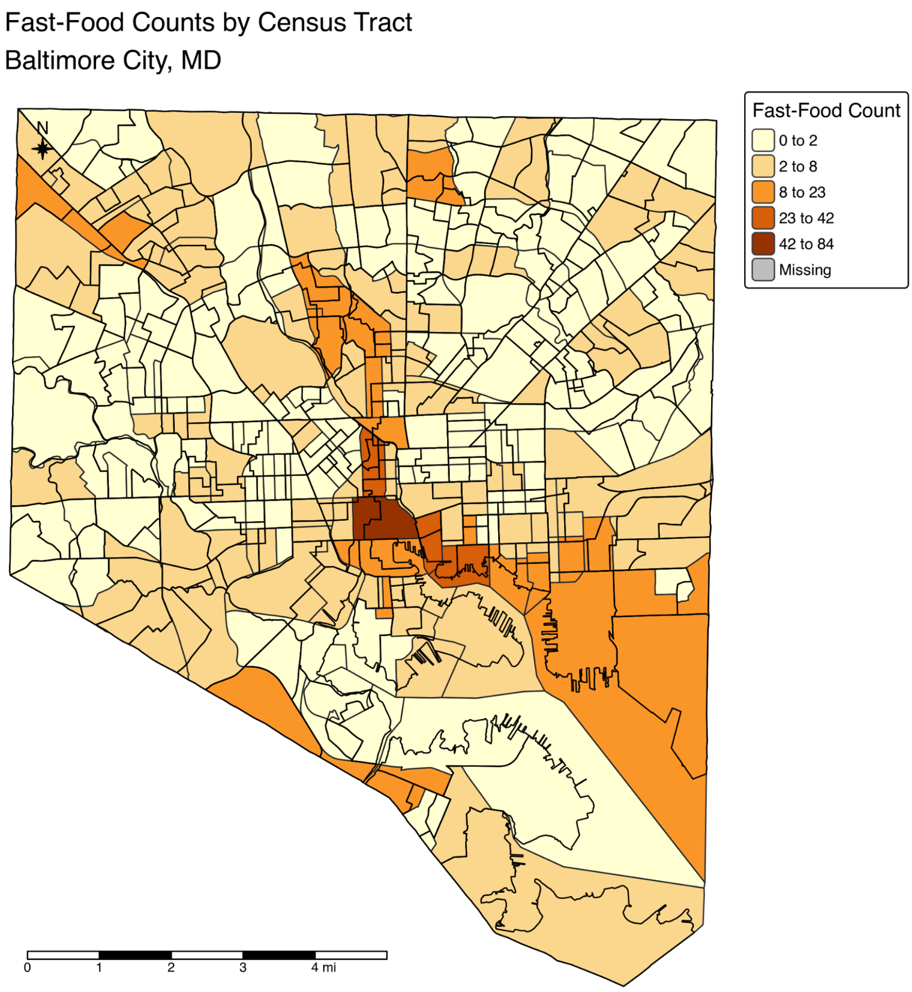
4. SaTScan Cluster Detection

Purely-Spatial Poisson Scan Clusters
Baltimore City, MD



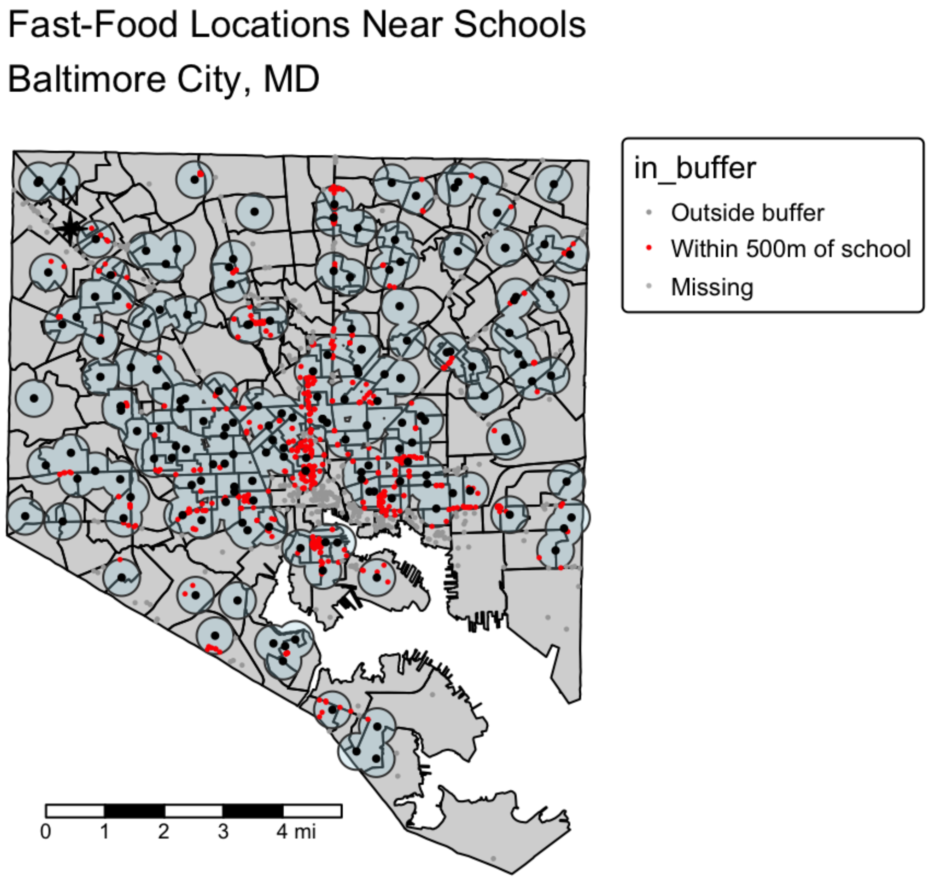
- Three statistically significant clusters identified ($p < 0.001$) through purely spatial Poisson scan
- Primary cluster located in central Baltimore with relative risk of 4.14 (301 cases vs. 72.77 expected)
- Secondary clusters found in eastern and western districts with relative risks of 2.08 and 3.66 respectively
- LISA analysis identified 11 census tracts in High-High clusters, confirming significant spatial concentration
- Clusters generally correspond with areas of higher population density and commercial activity
- Maximum concentration observed in a single tract with 84 fast-food locations, far exceeding the mean of 4.14 per tract

5. Fast-Food and Population Relationship



- Census tracts with higher population tend to have more fast-food establishments, suggesting demographic influence
- Fast-food counts range from 0 to 84 per tract (mean = 4.14), indicating highly uneven distribution
- Population-adjusted rates reveal different patterns than raw counts, with some lower-population areas showing disproportionately high fast-food density
- Relationship supports hypothesis that fast-food locations target areas of higher population concentration

6. School Proximity Analysis



Proposed Methodology for School Level Analysis

- Analyze fast-food clustering by school type (elementary, middle, high) using separate buffer zones
- Compare densities between different school levels to identify potential targeting patterns
- Measure distance relationships using multiple buffer rings (250m, 500m, 1000m)
- Test statistical significance of any observed differences between school types
- Control for confounding variables like population density and neighborhood characteristics
- Examine whether marketing strategies may differ based on student age demographics

Follow Up

- Analysis by different school types (elementary, middle, high) could reveal differential clustering patterns of fast-food locations by student age demographics
- Multi-ring buffer analysis (e.g., 0-100m, 100-500m, 500-1000m) would provide more detailed understanding of distance decay effects around schools
- The spatial analysis could be expanded by incorporating additional covariates into regression-kriging models
- SaTScan analysis could be enhanced by adjusting for additional covariates like income and population density
- Cross-K function analysis would provide more robust quantification of spatial relationships between schools and fast-food locations
- Targeted sampling in under-surveyed zones could reduce prediction uncertainty
- Validation against external health outcome data would strengthen public health relevance
- Use of additional environmental features (e.g., road networks, public transit) could improve contextual understanding of accessibility factors,
- MOST Definitely use Linear Reference Network